

Drought alters the structure and functioning of complex food webs

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Climate change is expected to make many regions of the world much drier over coming decades^{1,2}. More intense drought would transform rivers³ with potentially severe but largely unknown consequences at higher (multispecies) levels of organization⁴. Here we show experimentally how the intensification of drought may alter the underlying structure and functioning (biomass flux dynamics) of freshwater food webs—networks of species and their interactions⁵. Drought triggered substantial losses of species and links, especially among predators, leading to the partial collapse of the food webs. Total resource–consumer biomass flux was also strongly suppressed by disturbance, yet several network-level properties (such as connectance and interaction diversity) were conserved, driven by consumer resource fidelity and a substantial reconfiguration of fluxes within the webs as production shifted down the size spectrum from large to small species. Our research demonstrates that drier climates could have far-reaching impacts on the functioning of freshwater ecosystems.

Climate change is altering the global water cycle^{1–3}, with extensive impacts on local ecosystems^{4,6}. In fresh waters, future shifts in the distribution of water will alter river flows and create new hydrologic regimes³. Declining rainfall and overuse of water are predicted to increase the intensity and frequency of droughts in river systems³, threatening the structure and functioning of food webs, and their provision of valuable ecosystem goods and services⁴, such as biomass production⁶. Many species adapted to life in permanently flowing waters are vulnerable to drought^{7–9}, and declining, intermittent flows could modify habitats, stripping away sensitive species and size classes^{9,10}, with unknown but potentially powerful impacts on food web architecture and dynamics. Biomass fluxes among species are key to understanding such changes because they determine ecosystem production, nutrient cycling, and resilience of food webs⁶. In theory, disturbances may concentrate fluxes into a smaller number of species and feeding paths, eroding resilience to future change by removing alternative trophic pathways⁵. This could be mitigated, however, where compensatory mechanisms (for example, increased production of small taxa as large individuals are lost) reconfigure the food web sufficiently to maintain ecosystem functioning in the face of disturbance¹¹.

Despite the potential for drought disturbance to alter food web dynamics, previous research has focused on structural attributes of communities^{12–14} and little empirical evidence exists for predicting future change in functioning at these higher levels of organisation⁶. Most studies of stream drought have focused on predictable seasonal events and have not addressed how the novel intensified

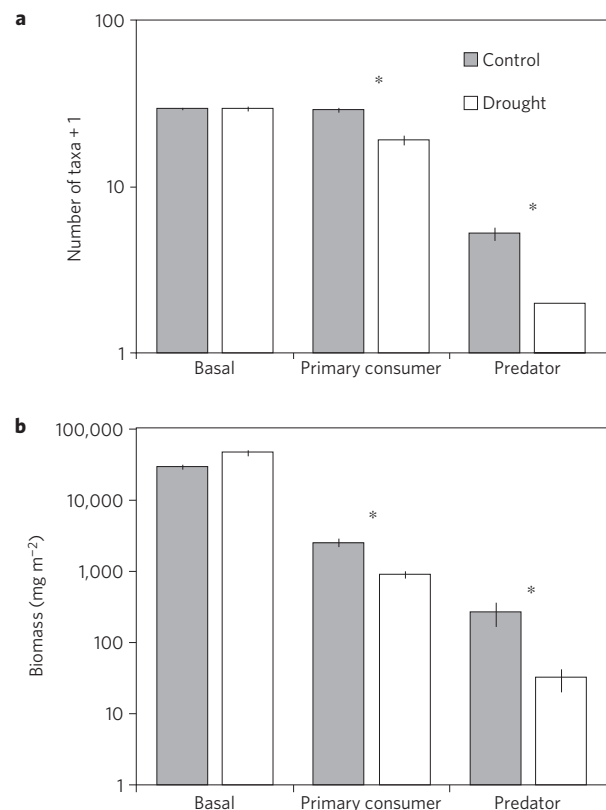


Figure 1 | Drought impacts on taxon richness and biomass varied among trophic groups. Mean (± 1 standard error of the mean, s.e.m.) taxon richness (**a**) and biomass (**b**) of basal resources, primary consumers and predators in drought and control food webs. Trophic group, drought treatment and their interaction significantly affected taxon richness and biomass (ANOVA, $n = 24$, $P < 0.0001$ in all cases). Asterisks above individual trophic groups denote significant differences between treatments (ANOVA, $n = 8$, $P < 0.05$).

droughts forecast by regional climate models may reshape stream ecosystems¹⁵. Here, we report the results of an experimental manipulation of highly taxonomically resolved quantified stream food webs conducted to measure the impacts of suprasedasonal drought on trophic structure and dynamics. We tested two hypotheses: first, that drought disturbance would generate turnover

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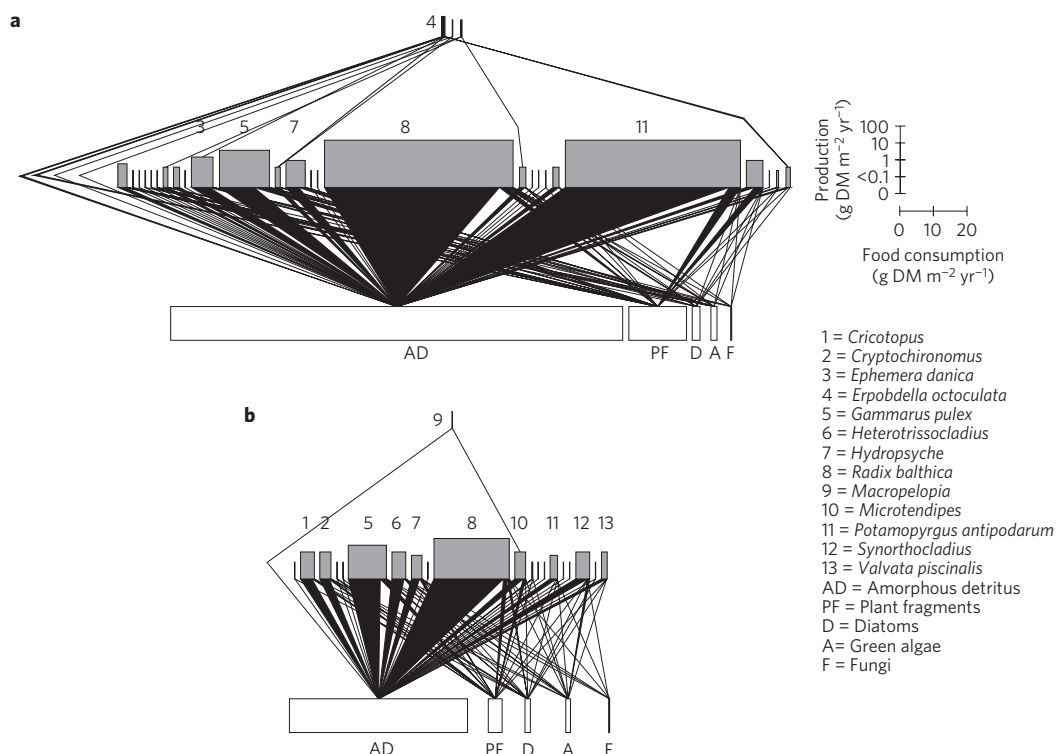


Figure 2 | Drought reconfigured stream food webs. Quantitative food webs in one block of control (**a**) and drought-disturbed (**b**) mesocosms (all webs shown in Supplementary Fig. S1). For each web, lower bars are basal resources, middle bars are primary consumers and top bars are predators. For each consumer, the height and width of the bars is proportional to mean annual secondary production and biomass flux from resources (total inflows to consumers), respectively. For basal species, the relative width of bars on the x axis is proportional to total consumption by invertebrates (total outflows from each resource to consumers), and for this trophic level production (y axis) was not quantified. The black triangles that link trophic levels illustrate the relative contribution of resource flows to the production of each consumer, summing to the total inflows. Numbers refer to consumer identity and letters distinguish categories of basal resource, omitting rare species (<1% total production). Flows from individual green algae and diatom taxa are grouped for display only. See Supplementary Tables S1 and S2 for full lists of taxa.

in biomass production, with small *r*-selected species replacing larger, longer-lived taxa with more *K*-selected traits¹⁶, and second, that dietary generalists would benefit over specialists¹⁷, modifying network properties and biomass flux.

Eight replicate mesocosms (stream channels 0.3 m × 12 m) fed by a river in southern England were subjected to intermittent flow (6-days of dewatering per month), mimicking suprasedasonal droughts^{8,9} that cause repeated stream drying (see Methods), or left as unmanipulated controls (under perennial flow), for two years. At the end of the experiment we examined treatment effects on taxon richness and biomass of benthic assemblages, and quantified network properties and biomass flux through the food webs. Food web nodes and links were identified by direct observation from benthic counts and gut contents analysis (3,643 individuals dissected), respectively. Biomass fluxes were quantified (g ash-free dry mass m⁻² yr⁻¹; ref. 18) using ingestion data and published estimates of secondary production¹⁹. Qualitative and quantitative food web metrics (for example connectance, linkage density) were used to characterize drought impacts on network architecture and fluxes²⁰. The webs, which are among the most highly resolved published to date, collectively encompassed 783 pairwise trophic interactions among 84 trophic elements, which consisted of detrital resources (leaf litter and amorphous detritus (that is, organic matter produced by biofilms)), fungi, benthic algae (green algae, diatoms and cyanobacteria; Supplementary Table S1) and a taxonomically diverse array of macroinvertebrate consumers (for example, amphipod shrimps, beetles, leeches, snails, worms and larval alderflies, caddisflies, mayflies, stoneflies and midges; Supplementary Table S2).

Drought significantly reduced the number of species (by 21%, from mean 61.0 ± s.e.m. 1.6 to 48.0 ± 1.5, Analysis of Variance (ANOVA) $F_{1,3} = 28.5$, $P < 0.05$) and links (by 34%, from 376.3 ± 36.2 to 248.0 ± 23.1, ANOVA $F_{1,3} = 21.77$, $P = 0.019$), although resilience to drought varied markedly among trophic groups (Figs 1 and 2). The taxon richness and biomass of basal resources was maintained (Fig. 1), despite a marked shift in dominance from encrusting green algae to diatoms²¹. In contrast, 37% of invertebrate primary consumer taxa were eliminated by drought and their biomass was suppressed by 64%. Extinction was most profound among the predators (78% loss of taxa, Fig. 1a), whose biomass collapsed by 88% under drought (Fig. 1b). For primary and secondary consumers, local extinctions were associated strongly with rarity (logistic regression, $\chi^2 = 9.398$, $P = 0.002$). A particularly striking result was that drought eroded food webs from the top-down, reducing proportions of predators and primary consumers relative to basal species. The loss of predators reduced mean (from 1.54 ± 0.004 to 1.39 ± 0.02, ANOVA $F_{1,3} = 31.58$, $P = 0.011$) and maximum (from 2.53 ± 0.05 to 2.16 ± 0.04, ANOVA, $F_{1,3} = 58.33$, $P = 0.005$) prey-averaged trophic level, consistent with theoretical predictions and field surveys along disturbance gradients^{13,14}. Drought caused high mortality among predators stranded in unfavourable habitat¹⁹. Potential food limitation was, however, partially offset by increased production of many small prey, notably chironomids, supporting our first hypothesis that *r*-selected species would exploit disturbance. This reflected a general shift away from larger taxa that had greater access to 'size refugia' from predation towards a more *r*-selected assemblage better able to access physical refugia from drought¹⁹.

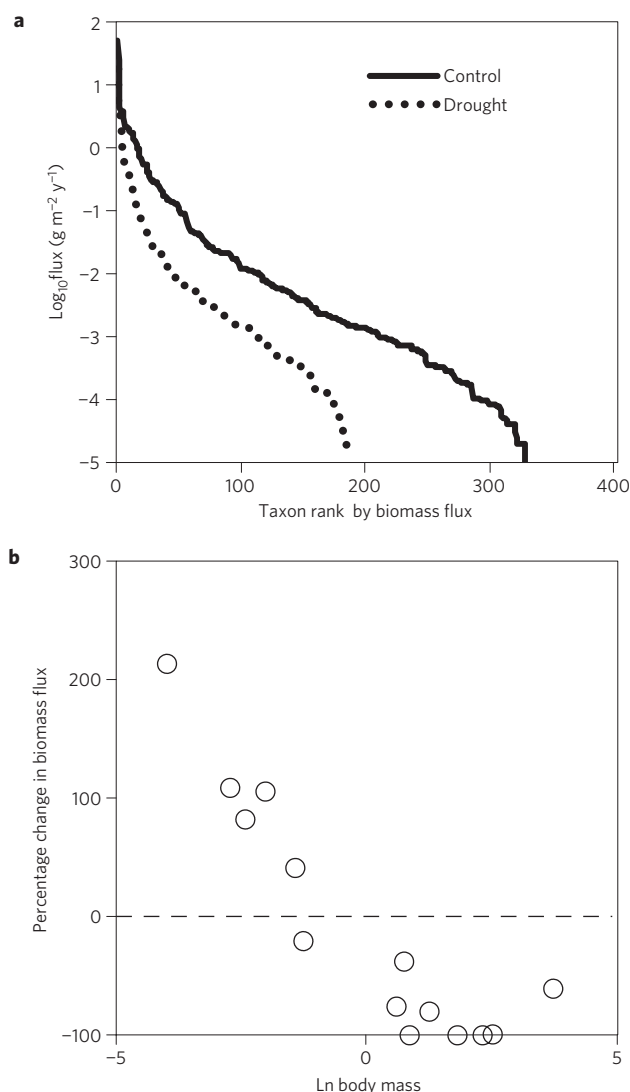


Figure 3 | Drought reduced and reconfigured biomass flux from resources to consumers. **a**, Magnitude of biomass fluxes in one block of disturbed and control assemblages (all webs shown in Supplementary Fig. S2). For each treatment, fluxes were ranked from left to right in order of decreasing magnitude. **b**, Shifts in biomass flux were related to body mass, as revealed by the relationship between mean body mass and mean change in biomass flux to consumers for strongest pathways (>1 g m⁻² yr⁻¹). Symbols above and below the dashed line in **b** denote taxa with increasing or decreasing fluxes in response to drought, respectively.

Food webs were largely detritus-based in both treatments, with biomass fluxes from detritus accounting for 96% of all flux (including to predators), and 90% of biomass flux was channelled through just 5% of links (Fig. 3). The largest fluxes were from amorphous detritus (that is, autochthonous organic matter produced by biofilms) to snails (*Radix balthica* (L.), *Potamopyrgus antipodarum* (Gray), mean 46% of flux to primary consumers). Only a small proportion of fluxes to primary consumers were transferred to predators (2.2%), because in both treatments many of the primary consumers (for example snails, caddis, shrimps) were too large (max. body mass 42 mg) for the gape-limited predators (max. body mass 32 mg) to handle. These size disparities between invertebrate predators and prey may effectively decouple their production in the webs²².

Drought strongly suppressed the total amount of biomass flux through the food webs by 59.8% (from 98.9 to 39.8 g m⁻² yr⁻¹;

Fig. 3a), despite extensive reconfiguration of fluxes among surviving species: that is, network restructuring did not fully maintain ecosystem functioning. Biomass flux increased through 43% of links, whereas drought caused steep reductions in the remaining pathways, including major fluxes to the highly productive snail populations (maximum 95% reduction of 43.6 g m⁻² yr⁻¹ to *Potamopyrgus antipodarum*). Shifts in biomass flux (that is, faster versus slower) to consumers were related to their body mass (logistic regression, $\chi^2 = 9.808$, $P = 0.002$), with increasing fluxes confined to small taxa such as midge larvae, and profound reductions for larger species, including snails, amphipod shrimps, caddis and mayfly larvae (Fig. 3b). This pronounced shift of production downwards through the size spectrum conserved the approximately log-normal relative distribution of fluxes within the webs (Fig. 3; equivalent interaction evenness among treatments, Table 1), precluding the potentially destabilising concentration of biomass within fewer species⁵.

Surprisingly, several structural network properties (linkage density, connectance and interaction diversity) were unaffected by drought (ANOVA, $P > 0.05$, Table 1, Supplementary Table S3) as were absolute and weighted measures of generality (number of resources per consumer) and vulnerability (number of consumers per resource). These five metrics are a manifestation of foraging behaviour and resource availability²³, and their constancy suggests consumer diets were consistent in the face of disturbance. Contrary to our second hypothesis, we found no evidence that disturbance promoted trophic generalists over specialists through indirect effects on food supply¹⁷, which would increase web connectance and linkage density. Rather, our data show drought acted directly by increasing consumer mortality through physiological stress¹⁹, with large species being most strongly affected, probably reflecting their relatively high metabolic demands²⁴.

Fresh waters are particularly vulnerable to climate change²⁵ and our results demonstrate that the present and future intensification of drought may have major effects on both biodiversity and ecosystem processes. The food webs underwent considerable restructuring in trophic height and the magnitude and distribution of biomass fluxes, driven by a reduction in consumer body-size, consistent with the emerging view that global change tends to have disproportionately strong negative effects on larger species¹⁰. However, the shift in biomass production from large to small species could not compensate fully for losses among the former, resulting in reduced overall biomass flux. Similar functional impairments of food webs can be expected in regions where future climate change exacerbates drought¹⁹. Such changes have implications for the future stability of food webs in an era of rapidly changing climate, as they could temper the effects of perturbations^{11,26}.

Methods

Experimental design and application. A drought experiment was conducted over 24 months (March 2000–February 2002) in four blocks of two linear stream mesocosms (width 0.33 m, length 12 m, depth 0.30 m) sited outdoors adjacent to, and fed by, a chalk stream at the Freshwater Biological Association River Laboratory (50°40′48″ N, 2°11′06″ W; refs 19,21; see Supplementary Method S1). The mesocosms replicate reaches of small headwater streams with stony beds and shallow subsurface sediments (20 cm depth)¹⁹. Following a two-month colonization period, an intermittent flow regime (6 days of flow cessation per month) was applied to one mesocosm in each block, mimicking supra-seasonal hydrologic drought^{8,9}. This simulated hydrograph approach allowed us to gauge how the biota responded to repeated stream drying caused by low rainfall²⁷ and/or overabstraction²⁸ as two predicted consequences of future climate change under Intergovernmental Panel on Climate Change (IPCC) scenarios^{1,2}. As with all mesocosm experiments, our flow manipulations necessitated some trade-off between realism and replication^{19,21}. Specifically, the simulated hydrograph may adequately capture the expected changes in the magnitude and frequency of river drying under climate change but does not necessarily mimic the expected changes in seasonality of these events. During

Table 1 | Quantitative weighted network properties were conserved under drought.

Metric	Control		Drought		ANOVA	
	Mean	s.e.m.	Mean	s.e.m.	$F_{1,3}$	P
Linkage density (LD_q)	5.94	1.16	4.20	0.64	3.33	0.165
Connectance (LD_q/S)	0.09	0.02	0.08	0.01	0.39	0.575
Generality (G_q)	1.82	0.11	1.59	0.08	3.89	0.143
Vulnerability (V_q)	10.06	2.23	6.81	1.21	3.29	0.167
Interaction diversity (ID_q)	3.98	0.35	3.41	0.32	3.36	0.173
Interaction evenness (IE)	0.47	0.03	0.43	0.03	1.90	0.262

6-day dewatering, surface flows ceased and exposed substrata dried in patches, although water remained in the interstices beneath the bed surface, and small pools persisted at intervals along the length of the dewatered channels^{19,21}. Surfaces of exposed substrata dried at natural ambient rates such that the stress experienced by organisms stranded in the mesocosms was consistent with those in adjacent drying stream reaches¹⁹. In the control mesocosms, flows were continuous throughout the experiment.

Sampling and processing. Estimates of invertebrate biomass were made from samples (Surber sampler 0.025 m², 300 µm mesh, $n = 3$ per channel) collected monthly from each mesocosm. Invertebrate secondary production was calculated from biomass, determined from 63,092 individual body length measurements, using the size-frequency method²⁹ (see Supplementary Methods S2), as part of a related study¹⁹. At the end of the experiment (after two years), we collected the entire macroinvertebrate assemblage in each mesocosm, using samples to construct food webs by direct observation of feeding links in 3,643 individuals in total, and determined biomass of basal resources (detritus and algae) as the ash-free dry mass of material collected from the surfaces of mineral substrata ($n = 8$) in each mesocosm²¹.

Food web construction. Binary food webs were constructed based on the presence/absence of resources in the diet of consumers. These webs were then quantified, with links expressed as flows of biomass from resources to consumers for each mesocosm community, using estimates of secondary production¹⁹. The trophic basis of production method was used to quantify directly-observed feeding links as biomass flux (F_{ij} , g m⁻² yr⁻¹) from resource i to consumer j (see Supplementary Method S3).

Qualitative and quantitative food web metrics. Metrics derived from binary webs based on the presence-absence of feeding links were: web size (S , the number of trophic elements in each web), number of pairwise feeding links (L), linkage density (L/S), directed connectance, the proportion of all possible links realized (L/S^2), generality (number of resources per consumer, L/S_{consumer}), vulnerability (number of consumers per resource, L/S_{resource}), and mean and maximum prey-averaged trophic level³⁰. Quantified food webs were compared using metrics derived from information theory^{5,20}, specifically quantified weighted measures of linkage density (LD_q), interaction diversity (ID_q), interaction evenness (IE_q), generality (G_q) and vulnerability (V_q) (see Supplementary Method S4).

Data analysis. Three-way analysis of variance (ANOVA) was used to test the effect of drought treatment, trophic group and experimental block on taxon richness and biomass (in SPSS 16.0). The analysis revealed a significant ($P < 0.05$) interaction between treatment and trophic position and two-way ANOVAs were performed subsequently to test for treatment and block effects on each trophic group. Treatment and experimental block were fixed factors in the analyses and data were log-transformed to homogenize variances. ANOVA also determined effects of treatment and block on food web metrics. Significance levels were adjusted for multiple tests using sequential Bonferroni correction. Logistic regressions tested for relationships between consumer extinctions (binary variable) and $\ln$ (abundance), $\ln$ (body mass), $\ln$ (body length) and voltinism (>1 , <1 cycles yr⁻¹), and between directional shifts in energy flux (increase or decrease under drought, binary variable) and $\ln$ (body mass), $\ln$ (body length) and voltinism. The presence-absence of consumer species in each web was determined from the collection of the whole community at the end of the experiment.

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Author contributions

M.E.L. was responsible for funding application, research design and planning, data collection and analysis, L.E.B. prepared and analysed food webs, F.K.E. collected and analysed data, A.M.M. applied for funds and planned the experiment, G.W. applied for funds and planned food web research. All authors discussed the results and wrote the manuscript.

Additional information

Supplementary information is available in the online version of the paper. Reprints and permissions information is available online at www.nature.com/reprints. Correspondence and requests for materials should be addressed to M.E.L.

Competing financial interests

The authors declare no competing financial interests.